

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q11-0017

High Point Fairfield 100 kV
March 18, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0017. The Interconnection Customer proposes a 160.2 MW generating facility interconnecting at the High Point Fairfield 100 kV substation in the Illinois area. The request was received on March 04, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 14 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$482,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The HIGH POINT FAIRFIELD-TROPIC ISLE 100 KV is overloaded to 102.7% of its 120 MVA emergency rating for the loss of the COMPO 230-1 kV XFMR. The project contributes 45.3% DFAX (72.6 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HIGH PT FAIRFIELD-TROPIC ISLE 100 KV	Reconductor line	\$482,000	100% (\$482,000)
	Total allocated to project		\$482,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.